Information Security Project: **50 marks (30 + 20, term paper + presentation)**
Weight: **10** (Aggregate Course Points)
Group Size: **2 or 3.**
CLO: **4**

Requirements: Write a *term paper* on a specific topic (chosen after approval from the instructor), and then *present* this topic in class towards the end of the semester. The term paper should describe the chosen topic in a **maximum of 6 pages**, with font size 11.It should reflect the understanding you have gained as a group on the topic, with references to the important research work done in that domain. The presentation should summarize your learnings, and you should be able to answer relevant questions at the end of the presentation.

If you want to choose another topic, you must first discuss this with me (via email or direct meeting). Only after due consideration, will I decide to accept or reject your chosen topic.

First Deadline: Inform the instructor of the group members and chosen topic by 8-Nov-2024 (11AM).

Next Deliverable: Term Paper (before presentation)

Final Deliverable: Presentation (last two weeks of classes)

Possible Topics:

Hasnain Lakhani (Senior Staff Software Engineer, Databricks. Previously, Engineering Manager, Facebook)

- **Static Analysis**: Improving the capabilities of static analysis for bug detection (perhaps using LLMs)
- **Fuzzing**: Studying how fuzzing performance can be measured more scientifically (there's a lot of research and open projects for LibAFL)
- **Privacy**: how do you reliably track fine grained properties of variables (for example, this variable is PII, Personally Identifiable Information, and must not be logged or go across regions)

Dr Hafiz Asif (Assistant Professor, Hofstra University, USA)

- **Fully homomorphic encryption**
- **Privacy-preserving computation**
- **Differential Privacy**

Dr Faisal Aslam (Assistant Professor, FAST-NU Lahore)

- **Zero-knowledge proofs**
- **Lattice-based cryptography**

- **Mathematical Attacks against RSA** (Chinese Remainder Theorem, N=p*q sieve)
- **Elliptic Curve cryptography**

Dr Zunera Jalil (Professor, Department of Cyber Security, Air University, Islamabad)

- **Phishing attack detection using AI**
- **OSINT for Law Enforcement Agencies** - for multilingual content
- **Fake news detection on social media**
- **Authorship attribution for forensics**

Others

- **Cybersecurity for Artificial Intelligence** ~ not AI for Cybersecurity (Shafqat Abbas, Cybersecurity Engineer, SkyElectric)
- **Quantum Key Distribution** (Aun Ali, BS-DS Graduate FAST-NU Lahore, 2024)
- **Quantum Resistant Cryptography** (Aun Ali) (Ref: 410.pdf (iacr.org))
- **Vulnerabilities of Quantum Computing** (Aun Ali)
- **Deep learning vulnerabilities against (textual) adversarial attacks** (Mumtaz)
- **Comparing security advice between Cybersecurity experts and LLMs** (Waleed Arshad, Current MS student at University of Wisconsin-Madison)
- **RF and UAV attacks** (M Huzaifah, Senior System Engineer, Sky Solutions)

This link may help you get started in searching for good research papers:

GitHub - prncoprs/best-papers-in-computer-security: This repo collects the best papers from top 4 computer security conferences, including IEEE S&P, ACM CCS, USENIX Security, and NDSS.